Self-critical rumination: A network approach

1. Introduction

Rumination is a repetitive thinking pattern about the causes, symptoms, circumstances and consequences of negative mood states and distress (Nolen-Hoeksema, 1991; Watkins, 2008). As a trans-diagnostic characteristic Nolen-Hoeksema & Watkins (2011), rumination could maintain and exacerbate existing psychopathology (Butler & Nolen-Hoeksema, 1994; Nolen-Hoeksema et al., 1999; Whisman et al., 2020). In general, it contributes to the onset or recurrence of several symptoms and disorders (Aldao & Nolen-Hoeksema, 2012; Lyubomirsky et al., 2015; Lyubomirsky & Nolen-Hoeksema, 1993; Nolen-Hoeksema et al., 2008; Ruscio et al., 2015; Watkins & Roberts, 2020; Whisman et al., 2020), examples entail major depressive disorder (Broderick & Korteland, 2004; McLaughlin et al., 2014; Nolen-Hoeksema et al., 2008; Wilkinson et al., 2013), generalized anxiety disorder (Nolen-Hoeksema, 2000), post-traumatic stress disorder (Moulds et al., 2020), bipolar disorder (Kovács et al., 2020), borderline personality disorder (Selby et al., 2008), social phobia (Abbott & Rapee, 2004), or psychosis (Hartley et al., 2013).

Rumination is often operationalized as depressive rumination and consequently measured by the *Response Style Questionnaire*, or its shorter form the *Ruminative Response Scale* (Nolen-Hoeksema, 1991; Treynor, 2003). Two subscales within RRS: the *brooding* and *reflective pondering* (or *reflection*) were identified (Treynor, 2003). Brooding is a maladaptive, passive dwelling on negative events and associated affects, while reflection is a more adaptive (or less pathological) form of thinking, a non-evaluative introspection towards one's mood. Several empirical findings support the adaptive-maladaptive distinction of brooding and reflection. Brooding, but not reflection, increased the risk of substance use among people with history of illicit drug use (Memedovic et al., 2019), and was associated with attentional disengagement difficulties (Allard & Yaroslavsky, 2019). Apart from brooding and reflection, many other rumination types or facets were described. For example, people, especially those with social phobia, tend to ruminate about their mood after a social event or performance (Abbott & Rapee, 2004). Additionally, individuals may dwell upon infuriating memories or on anger itself (Bushman et al., 2005; Pont et al., 2017). Furthermore, repetitive thinking can also target feelings of sadness (Peled & Moretti, 2009).

These results indicate that not just the style of thinking, but the content of these ruminative thoughts has special relevance (Smart et al., 2015). More recently, research has started to investigate the role that rumination plays in maintaining high levels of self-criticism (Kolubinski et al., 2017; Smart et al., 2015). Self-criticism is a form of negative self-evaluation with one harshly and hostilely judging oneself, especially in the context of mistakes, failures, unreached goals or unmet standards (Rose & Rimes, 2018; Shahar et al., 2014). Self-criticism was found to be a vulnerability factor in many forms of psychopathology such as depressive and anxiety disorders, binge eating or non-suicidal self injury (Glassman et al., 2007; Rose & Rimes, 2018; Shahar et al., 2014). Rumination was found to be a significant mediator between self-criticism and depression across 2.5 years (Spasojević & Alloy, 2001), and between self-criticism and suicidal ideation over 3 months (O'Connor & Noyce, 2008). Thus, we can assume that rumination is one of those processes by which these negative self-devaluating thoughts are sustained and connected to psychopathology. Investigating rumination in the context of self-critical thoughts is paramount, especially in light of the results of a recent meta-analysis which indicated that the harsh and persistent forms of self-criticism did not change through short-term interventions (Werner et al., 2019).

Self-critical rumination significantly differs from depressive rumination, mostly by the focus of thoughts. While in self-critical rumination, individuals concentrate on their lack of self-worth or those aspects of the self of which they are ashamed of (Smart et al., 2015), in depressive rumination one's attention is directed towards the symptoms of depression (Nolen-Hoeksema, 2000). Indeed, in a recent study, self-critical rumination was strongly related to shame and predicted sensitivity to stress after a perceived failure (Milia et al., 2020).

Positive meta-cognitions in self-critical rumination (Kolubinski et al., 2017) likely has less importance than in depressive rumination (Papageorgiou & Wells, 2001). One can assume that in depressive rumination the positive, while in self-critical rumination the negative meta-cognitions (e.g. "*I will get depressed if I don't stop reviewing my self-critical thoughts*") are those processes which maintain the ruminative process (Kolubinski et al., 2017). The Self-Critical Rumination Scale (SCRS, Smart et al. (2015)), developed to assess the construct, showed incremental validity over other rumination scales (including the widely used RRS and RSQ) and self-criticism scale in explaining interindividual differences in borderline personality features and general distress. However, in the same study, SCRS and RRS were highly correlated, indicating that those individuals who ruminate in a more general or broader way are also likely to ruminate on numerous topics, including self-devaluating thoughts and vice versa. Previous work also supported the notion that individuals can show different forms of repetitive thought and they can ruminate and worry simultaneously, or they are able to passively brooding on something and adaptively reflect on other topics (Watkins, 2008).

By considering the types of rumination (brooding vs. reflection) and their content (such as self-critical thoughts, anger, performance in social situations), we can identify several distinct forms of rumination.
Regarding the challenge of defining boundaries between various subtypes, adopting a network approach becomes valuable for analyzing and understanding their interconnections. Modeling rumination as activation in a system of interacting components, such as a network would allow us to shift away from the distinctive labeling of brooding, relfection or self-critical aspects of rumination and would allow a more detailed, pattern-like view on the psychology of repetitive thoughts (see also at Bernstein et al., 2019). In this network approach, items are not assumed to have a common, underlying cause, nor can we expect that resulting networks fully map and cover the whole construct of trait rumination. Psychological networks such the ones presented below rather can be used to model components (items, and in our case, constructs measured by different scales or subscales) in dynamic interaction and to test past distinctions and classifications of rumination in a data-driven way. Contrary to latent variable models, network analysis does not seek for a common cause of co-occurrence of indicators (e.g. different items of a questionnaire) but shift the focus toward the relationships and interactions between variables, items or other constructs of interest (Epskamp, Kruis, et al., 2017). As opposed to *formative* models where observed scores define an attribute, or *reflective* models where a latent attribute is causing observables; in this alternative approach, psychological attributes (such as rumination) are conceptualized as *network* of directly related observables (Schmittmann et al., 2013). *Psychological networks* have their own characteristics which differentiate them from other kinds of networks of graph theory (Epskamp, Borsboom, et al. (2017b); Epskamp & Fried (2018); Newman (2010)). Vertices (also called nodes) usually represent some psychological measures or items of psychological surveys or questionnaires, while the edges of the network reflect the corresponding estimates of association derived from the data. Thus in a network model, the construct of rumination could be interpreted as a system in which some aspects relate or even stimulate each other, while other aspects have negative or inhibiting effects (Costantini et al., 2015). With this data-driven approach we can test how self-critical and depressive ruminative processes or the different items of the ruminative questionnaires cluster and relate to each other and we can gain important insights which could remain hidden by relying only on latent variable models (Bernstein et al., 2019). Having two rumination-related questionnaires, the Ruminative Response Style (*RRS*) and the Self-critical Rumination Scale (*SCRS*), we assumed the following hypotheses:

(1.) Network analysis will reveal their respective internal structure previously established in the literature (such as *RRS* having a two-factor structure: *reflection* and *brooding*; while *SCRS* will show a single-scale structure); (2.) on the other hand, we hypothesized that in their common network, items overlapping in scope will change these clear outlines. According to our hypothesis, items of the *RRS* scale *brooding* will have statistically significant association with the self-critical aspects of rumination; (3.) we also assumed a combined rumination network, representing the generic (more trait-like) and the self-critical aspects of rumination will exhibit a three-component structure.

Previously, network modeling was successfully applied to investigate depressive rumination by testing the original 22-item version of RRS (Bernstein et al., 2019), but to the best of our knowledge no other rumination questionnaires were investigated, especially in terms of self-critical thoughts. We provide a glossary of the main network analysis related concepts in Table 8. of the Appendix.

2. Methods

2.1 Participants and procedure

Students from several Hungarian high schools participated in the study on a voluntary basis. The first stage of the recruitment process was to contact the heads of the schools to acquire institutional consent. Afterwards, classes were visited asking the students to participate. Since, the participants were under-aged, parents as well as students were issued with written informed consent letters. Following their consent, students were asked to fill out paper-pencil questionnaires in their own classroom within one class session. Participation in the study was anonymous and voluntary. Data collection was supervised by trained undergraduate students of psychology. No teaching staff were present. The total sample of the study comprises data from 851 participants (aged 14-19 years, females= 502 (59%), males=349 (41%) collected via convenience sampling method. The mean age was 16.1 (SD=1.1) years. The protocol was designed in accordance with the Declaration of Helsinki and the study was ethically approved by the Institutional Review Board (note: the name of the university is masked for review process).

2.2 Measures

The *Self-Critical Rumination Scale*; (Smart et al., 2015) was used to assess those thoughts that criticize the self for perceived failures, mistakes or bad habits in a prolonged and repetitive manner. The SCRS is a self-reported, one factor scale that consists of 10 items. Participants were asked to rate the statements (i.e. "*My attention is often focused on aspects of myself that I'm ashamed of*"; "*I criticize myself a lot for how I act around other people*") on a 4-point Likert scale from 1 (not at all) to 4 (very much). The scale showed high internal consistency in the original version (α=0.92). We obtained a satisfactory estimate (α=0.89).

The short form of *Ruminative Response Scale*; (Treynor, 2003) was used to measure mood dependent rumination, which consists of 10 items and quantifies ruminative thoughts via two dimensions (each composed of 5 items). The *brooding* subscale measures the maladaptive form or ruminative thinking (i.e. "*What am I doing to deserve this?*"), while the *reflective pondering* (or reflection) factor measures a more adaptive form of self-reflecting thinking (i.e. "*I write down what I am thinking and analyze it*"). Participants are asked to indicate on a 4-point Likert scale how often the items apply to themselves when they feel down, sad or depressed (1=almost never; 4=almost always). Internal consistency was found to be satisfactory for RRS (α=0.75), as well as for the *reflection* (α=0.70) and *brooding* (α=0.65) scales. Besides the SCRS and RRS, basic socio-demographic information (e.g. age, gender, parents' level of education) was recorded. See full item information of *SCRS* and *RRS* in Table 1.

Table 1: Measures

Questionnaire	Scale	Item	Label
RRS (1)	Brooding	Think ‘What am I doing to deserve this?’	deserve
RRS (2)	Reflection	Analyze recent events to try to understand why you are depressed	analyze
RRS (3)	Brooding	Think ‘Why do I always react this way?’	react
RRS (4)	Reflection	Go away by yourself and think about why you feel this way	alone1
RRS (5)	Reflection	Write down what you are thinking and analyze it	write
RRS (6)	Brooding	Think about a recent situation, wishing it had gone better	wish
RRS (7)	Brooding	Think ‘Why do I have problems other people don’t have?’	problem
RRS (8)	Brooding	Think ‘Why can’t I handle things better?’	handle
RRS (9)	Reflection	Analyze your personality to try to understand why you are depressed	person
RRS (10)	Reflection	Go someplace alone to think about your feelings	alone2
SCRS (1)	Self-critical	My attention is often focused on aspects of myself that I’m ashamed of.	aspects
SCRS (2)	Self-critical	I always seem to be rehashing in my mind stupid things that I’ve said or done.	said
SCRS (3)	Self-critical	Sometimes it is hard for me to shut off critical thoughts about myself.	thoughts
SCRS (4)	Self-critical	I can’t stop thinking about how I should have acted differently in certain situations.	situation
SCRS (5)	Self-critical	I spend a lot of time thinking about how ashamed I am of some of my personal habits.	habits
SCRS (6)	Self-critical	I criticize myself a lot for how I act around other people.	act around
SCRS (7)	Self-critical	I wish I spent less time criticizing myself.	mistakes
SCRS (8)	Self-critical	I often worry about all of the mistakes I have made.	different
SCRS (9)	Self-critical	I spend a lot of time wishing I were different.	time
SCRS (10)	Self-critical	I often berate myself for not being as productive as I should be.	productive

Note. Questionnaires with their scales and full item texts. Tag is the abbreviated tag name used for a specific item in the present paper.

2.3. Network modeling, descriptives and communities

In the current study, we constructed a dense, undirected network of rumination-related psychological constructs. The vertices represent the items of the *RRS* and *SCRS*, while edges are penalized partial correlations using LASSO (Least Absolute Shrinkage and Selection Operator) penalization (Friedman et al., 2007; Tibshirani, 1996).

We used individual and combined networks of *RRS* and *SCRS* with respective penalized correlation matrices using the *EBICglasso* function of the *qgraph* package (Epskamp et al., 2012). See our Git repository for other parameter specifications. In order to shrink the edges, achieve high specificity, and get a less dense network, we let the function to exclude any elements of the underlying precision matrix below a threshold value (*threshold* parameter set to TRUE).

The LASSO uses *L1* (absolute) regularization (Friedman et al., 2007; Tibshirani, 1996). In the current study, the method resulted in a partial correlation network where edges represent direct associations between pairs of items, while the influence of all other items in the network are kept controlled. The regularization by *L1* penalty lets trivially small partial correlations to be set to zero (McNally et al., 2017), thus ensure a *sparse* network which are more interpretable (Epskamp & Fried, 2018).

The tuning parameter of *gamma* in the graphical LASSO algorithm has been set to 0.5 value in accordance with (Bernstein et al., 2019; McNally et al., 2017). The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

We applied the *goldbricker* function from the R package *networktools* (Jones, 2017) in order to search for pairs of items in the *RRS*, *SCRS* and their combined networks that were highly correlated ($r > 0.75$) and exhibited the same pattern with other items. After applying the LASSO regularization, no further dimensionality reduction was necessary based upon the *goldbricker* evaluation.

A comparative overview of the three networks was produced with their descriptives (degree distribution) and their centrality measures (strength, closeness, betweenness; Kolaczyk (2020)), and their One- and Two-Step Expected Influence (EI) measures to understand the cumulative importance of vertices (Bernstein et al., 2019; Donald J. Robinaugh, 2016; Heeren et al., 2018). In order to further elaborate the question of vertex importance, we identified *bridge nodes* using the functions provided by the *networktools* R package (Jones, 2017) to identify bridge vertices, in other words, connectors between sub-graphs (see an example at Bernstein et al., 2019). We estimated both one-step and two-step bridge expected influence measures which reflect the sum of edge weights connecting one vertex to other groups on vertices (or communities); and also express a vertex's secondary influence on the rest of the network, respectively. Higher influence scores indicate that a certain vertex is more likely to active neighbouring vertices or communities. In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. Also, to get an understanding of communities of vertices and to see if vertices can be grouped into neighbourhoods corresponding to the previously established sub-scale structure of *RRS* and *SCRS*; and to investigate their separation in their combined network, we used the community detection methods available in the *igraph* R package (Kolaczyk, 2020). When detecting possible communities of vertices, we followed guidelines by (Bernstein et al., 2019; Kolaczyk, 2020; Reichardt & Bornholdt, 2006). In order to mitigate the effect of choice of community detection algorithms, we performed more available algorithms from the *igraph* package multiple times (B=1000) in weighted and unweighted settings, too.

2.4 Internal Structure with an exponential random graph model

In order to measure the impact of of one item belonging to a questionnaire (*RRS*, *SCRS*) and their respective scales (*brooding*, *reflection* and *self-critical*) on the combined network structure, we constructed an *exponential random graph model* (ERGM) (Hunter, 2007). In our model, *endogenous* predictors (related to the nature of the network structure) included the number of edges in the network (*edges* member), and a generalization of triadic structures based on alternating sums of *k-triangles* (Goodreau et al., 2009; Hunter, 2007) to involve some higher-order global network structures. This predictor served as a reflection on clustering in the form of completed triangles of vertices. As estimator, we used *GWESP* (*geometrically weighted edgewise shared partner distribution*) (see Goodreau et al., 2009). *GWESP* has the advantage of taking a parametric form of the shared common vertex count distribution on each edge, giving each connecting vertex a declining positive probability of two selected vertices being connected. In order to capture *exogenous* effects (i.e., vertex attributes), we added *questionnaire* and *sub-scale* as predictors, especially testing for *homophily* (assuming that items belonging to the same questionnaire and its scale will have higher probability of being connected by an edge).

2.5 Association Network Inference

In order to understand the nature of the edges in the combined association network, we conceptualized the combined network as an *association network*, associations given by the correlation coefficients between responses to items. Our aim was to discern real edges from non-edges (when two items are not correlated) and derive the mixture null distribution in a data-driven fashion. The R package *fdrtool* (Strimmer, 2008a; Strimmer, 2008b) allowed us to model the probability density function and estimate η_0 (the fraction of true null hypotheses); and (1-eta_0) for the true alternative hypotheses. The mixture model for some statistic S thus takes the following form (where in our analysis, S is the partial correlation coefficient).

$$f(S) = \eta_0 f_0(S) + (1 - \eta_0) f_1(S)$$

To apply adjustment due to the number of performed tests, we controlled p-value by the *False Discovery Rate* (*FDR*; the rate of false rejections against the number of rejections).

2.6. Latent Network Modeling

We constructed an *eigenmodel* to further explore the structure of the combined graph of *RRS* and *SCRS* (Hoff et al., 2002; Kolaczyk, 2020; methodology following Mair, 2018) and how well the network structure predicts edges. Our expectation was that sub-scales will have non-trivial impact on the resulting eigenvalues. In our two-part analysis, first we used eigenvalue decomposition in order to use the first two eigenvectors reflective to our scale- and sub-scale structures of *RRS* and *SCRS*. In the second part, we defined three conditions of our *eigenmodel*. In the first condition we used the original *eigenmodel* (discussed above) without any additional regression components, defining the adjacency matrix only as a function of the pairwise latent effects summarized in the following matrix.

$$Y_{ij} = f(u_i \Lambda u_j)$$

In the second and third conditions, we defined optional regression components in addition to the pairwise latent effects, where $\mathbf{x}_{ij}$ is a pair-specific predictor vector. This extends our model to the following equation.

$$Y_{ij} = f(\beta' x_{ij} + u_i \Lambda u_j)$$

In the second condition, we cross-validated an extended *eigenmodel* with *Questionnaire* (*RRS* or *SCRS*) serving as grouping factor of vertices (the model captured the pairwise connections and Questionnaire playing an additional role in the edge formation process). In the same manner, the third condition tested a model with the *Scales* of *brooding*, *reflection* and *self-critical* serving as pair-wise covariate predictor effects.

2.7. Power analysis, network accuracy and stability

Accuracy and stability have an important role in estimating network parameters when samples might vary in characteristics and size (Epskamp, Borsboom, et al., 2017a, 2017b). One way of estimating network accuracy is to produce a 95% confidence interval of edge-weights with some variant of bootstrap. In the present paper, we chose the non-parametric bootstrap estimate for interpretability. Network stability is quantified with the *Correlation Stability* (*CS*-coefficient): an estimate of the maximum proportion of cases that can be dropped (with 95% certainty) to retain a correlation with the original centrality higher than some fixed threshold value (in the present paper, 0.7). Simulation studies (Epskamp, Borsboom, et al., 2017b) lead us in our expectation as the *CS*-coefficient should be between above 0.25, preferably higher than 0.5.

3. Results

3.1 Descriptives, network descriptives and community detection

Pearson correlation coefficients and Cronbach's αs of the scales, means and standard deviations (for the total sample, and for females and males separately) are presented in Table 1. below. The data being skewed and homoscedasticity violated between gender groups, we estimated 20% trimmed Cohen's d coefficients. With medium-to-high effect sizes, Cliff's delta suggests group differences for female participants having higher scores on *RRS*, *SCRS* and RRS sub-scales (with at least 0.577 or higher probability). As expected (Smart et al., 2015), significant strong positive correlations were found between SCRS and RRS total scores (and both sub-scales), indicating that self-critical rumination somewhat differs from depressive mood related rumination, but people who are likely to ruminative self-critically are also likely to ruminate about their mood.

Table 2: Sample and Scale Descriptives

	1.	2.	3.	4.
Correlations:				
1. RRS				
2. RRS - brooding	0.82			
3. RRS - reflection	0.85	0.40		
4. SCRS	0.64	0.63	0.44	
Female Mean (SD)	21.59 (4.85)	10.88 (2.77)	10.71 (3.08)	25.19 (7.15)
Male Mean (SD)	19.07 (4.59)	9.65 (2.72)	9.42 (2.80)	21.23 (6.71)
Cronbach's α	0.75	0.65	0.70	0.89
Cohen's d (20% trimmed)	0.61	0.54	0.47	0.53
Cliff's delta	P(f>m)=0.64	P(f>m)=0.59	P(f>m)=0.58	P(f>m)=0.62

Note. *RRS= Ruminative Response Scale. SCRS= Self-Critical Rumination Scale. SD=standard deviation. P(f>m)=probability of difference of female scores over male. All correlations are significant at p<.001.*

The undirected, weighted, associational networks of *RRS*, *SCRS* and their combined structure can be seen below (see Figure 1). All three networks proved to be connected (any vertex is reachable from every other). Comparing diameters to mean distances, we see densely connected graphs (for the *RRS* network, its diameter is 0.467 compared to its mean distance of 0.207; the *SCRS* and the combined networks show similar ratio patterns). On the plot, vertices represent the corresponding items of each questionnaire with labels showing item number, sub-scale and a label for each item. Sub-scales are presented in a colour-coded fashion as the outer circle of each vertex. Edges are the *LASSO L1*-penalized partial correlation coefficients.

The *RRS* (order of 10, size of 17) and *SCRS* (order of 10, size of 16) networks are densely connected networks. Analyzing their structure, while *SCRS* exhibits no *articulation point* [*cut vertex* whose removal would disconnect the network; Kolaczyk (2020)]. The vertex of the *'problem'* (*"What am I doing to deserve this?"*) item of *RRS* acts as an *articulation point*, as it would cut off the *'deserve'* (*"Why do I have problems other people don't have?"*) vertex from the rest of the network. Articulation points are marked with * on Figure 1. Besides *size* and *order*, on Figure 1. we also indicated the *Clustering coefficient* (*transitivity*), the *diameter* and the average vertex-to-vertex path lengths (*mean distance*).

The combined network (order: 20, size: 39) has no articulation points. The previously identified articulation point of the *RRS* network (*problem* vertex) is no longer a cut-off vertex when all rumination items (brooding, reflection and self-criticism) are encompassed in one graph.

In the *RRS* network two Reflection items proved to be have highly positive influence on their neighbourhoods, these are the 'Go away by yourself and think about why you feel this way'- [Betweenness=6.00; Exp. Inf.=1.53; Two-Step Exp. Inf.=1.63] and the 'Analyze your personality to try to understand why you are depressed'- [Betweenness=14.00; Exp. Inf.=1.26; Two-Step Exp. Inf.=0.97] items. The item 'Think 'Why can't I handle things better?''- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA] was the most influential item of the Brooding scale in *RRS* network. The items with the most negative, prohibiting influence of the *RRS* graph were 'Think 'What am I doing to deserve this?''- [Betweenness=0.00; Exp. Inf.=-1.09; Two-Step Exp. Inf.=-1.22]; 'Analyze recent events to try to understand why you are depressed'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA] and 'Write down what you are thinking and analyze it'- [Betweenness=0.00; Exp. Inf.=-1.26; Two-Step Exp. Inf.=-1.05]. These items stand out in their characteristic that their two-step influence metrics (influence on the rest of the graph also taken into account) are negative.

In the *SCRS* graph highly positive influence were contributed to items such as 'I spend a lot of time thinking about how ashamed I am of some of my personal habits.'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA] and 'I wish I spent less time criticizing myself.'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA]. The highest negative influence items of *SCRS* were 'I can't stop thinking about how I should have acted differently in certain situations.'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA] and 'I spend a lot of time wishing I were different.'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA].

In the combined *RRS* - *SCRS* graph, the item of 'Go away by yourself and think about why you feel this way'- [Betweenness=NA; Exp. Inf.=NA; Two-Step Exp. Inf.=NA] not only proved to be highly positively influential, but also increased its Betweenness centrality, meaning it started to act as a connecting point between sub-domains of the network. The rest of the most positively influential nodes of this network are from *SCRS* such as 'I can't stop thinking about how I should have acted differently in certain situations.'- [Betweenness=28.00; Exp. Inf.=1.61; Two-Step Exp. Inf.=1.47], 'I spend a lot of time thinking about how ashamed I am of some of my personal habits.'- [Betweenness=12.00; Exp. Inf.=1.47; Two-Step Exp. Inf.=1.70], 'I criticize myself a lot for how I act around other people.'- [Betweenness=51.00; Exp. Inf.=1.06; Two-Step Exp. Inf.=1.15] and 'I often worry about all of the mistakes I have made.'- [Betweenness=38.00; Exp. Inf.=1.32; Two-Step Exp. Inf.=1.16]. With regard to the negative influence of network activation in the combined graph, the pattern is dominated by *RRS* items such as 'Think 'What am I doing to deserve this?''- [Betweenness=0.00; Exp. Inf.=-1.82; Two-Step Exp. Inf.=-1.80], .'Think 'Why do I always react this way?''- [Betweenness=15.00; Exp. Inf.=-1.23; Two-Step Exp. Inf.=-1.33], 'Write down what you are thinking and analyze it'- [Betweenness=11.00; Exp. Inf.=-1.12; Two-Step Exp. Inf.=-1.11], 'Think about a recent situation, wishing it had gone better'- [Betweenness=3.00; Exp. Inf.=-0.81; Two-Step Exp. Inf.=-0.78] and 'Think 'Why can't I handle things better?''- [Betweenness=7.00; Exp. Inf.=-1.08; Two-Step Exp. Inf.=-1.23]. Note the item of *write* increased its Betweenness compared to the standalone *RRS* graph. Among the *SCRS* items, the 'I wish I spent less time criticizing myself.'- [Betweenness=2.00; Exp. Inf.=-0.94; Two-Step Exp. Inf.=-0.65] was the most negatively influential on the common network.

In order to analyze their respective internal structures, we performed community detection using the *igraph* package in R. On Figure 1., the inner dot of each node provides colour coding for our community detection results in a manner that they correspond to the True Positive Rates (TPR) of each sub-scale the item belongs to. In other words, darker shaded nodes have higher average match between the detected community (the communtiy found by the algorithm) and the original subscale membership [TPR ranging between 0 and 1]). Lower TPR in this sense means that the item had on average lower rate of being member of its subscale community.

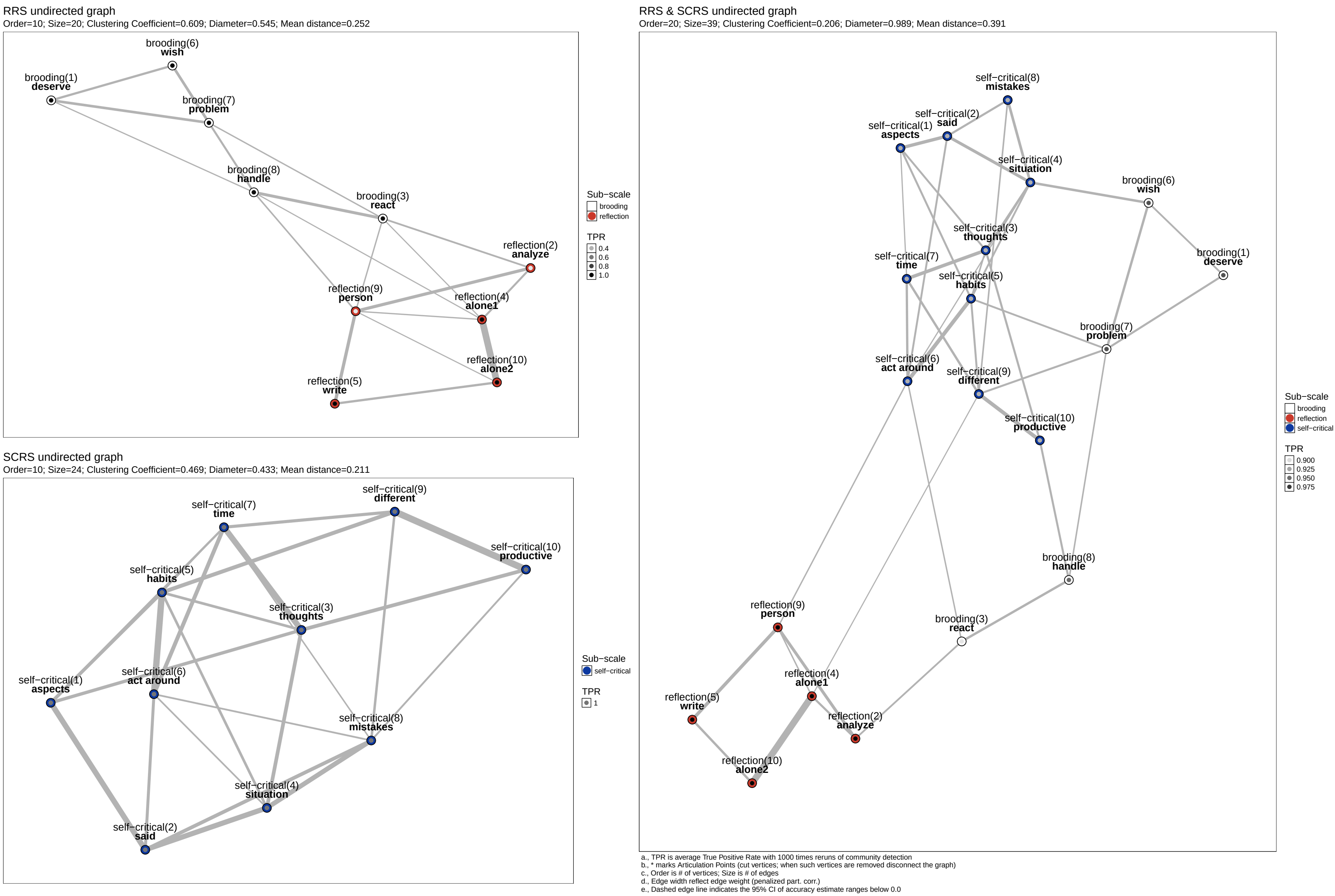

Figure 1: Tableau view of three graphs of rumination

As the choice of algorithm has non-trivial impact on the identified community structure, after having selected the *"optimal"*, the *"spinglass"*, the *"walktrap"* and (to also include hierarchical clustering) the *"fast greedy"* methods, we tested each 1000 times on the three networks. We also provided an edge weight vector for community detection in the form of the *LASSO*-penalized partial correlation coefficients (or alternatively, omitted weights). Presented TPR values are therefore averages of estimates from all these conditions.

The *RRS* two-factor structure could be recreated with all four algorithms, the inclusion of the weight vector not having effect on sub-scale membership. For the sub-scale of *brooding* the average *true positive rate* (TPR) was 0.8; for *reflection* 0.743. In case of the *reflection* sub-scale, lower average TPR becomes tangible with the inspection of items 2 (*"analyze"*; *"Analyze recent events to try to understand why you are depressed"*) and 9 (*"person"*; *"Analyze your personality to try to understand why you are depressed"*) whose low TPR indicates that depending in the choice of community detection they have a chance of forming their own community or being classified as members of the *brooding* scale, rather than being part of *reflection*. The one-factor structure of the *SCRS* has been reestablished.

The combined rumination network shows a clear expected structure, with having high TPR for each sub-scale (on average, *brooding* (0.9), *reflection* (1) and *self-critical* (0.8)). Commenting on the stability of the network community structure, the *RRS reflection* sub-scale exhibits strong TPR rates, and the weakness of the two items from their respective stand-alone network for items 2 and 9 are no longer present in the combined graph. The *RRS brooding* scale has good separation in terms of TPR, except for its item 3 (*react*) having <0.9 TPR (*"Why do I always react this way?"*). The one-factor SCRS has also a separable community with high (>0.925) TPR. The combined network has large TPR of 0.9; while keeping only the *spinglass* algorithm (as used by Bernstein et al., 2019), we get high average 1.0 TPR for the combined graph.

Please refer to our vertex summary table in the Appendix (Table 4.) for actual descriptives, including expected influence, bridge node related statistics and average True Positive Rates for each vertex. Where applicable, we used the standardized scores.

Edges are remaining penalized partial correlations between items, after LASSO-penalization. As noted at Figure 1., dashed edges are where the 95% confidence interval of accuracy estimates entails 0.0, thus indicating that the two nodes of the given edge might be conditionally independent (i.e. having no association), given sample characteristics change. Such edges are present in the RRS network, and the SCRS network respectively (the two left graphs in Figure 1), and also in the combined graph. One notable, inter-questionnaire edge which has high, non-zero accuracy is between items *wish* of the RRS Brooding (*Think about a recent situation, wishing it had gone better*) and *situation* of SCRS (*I can't stop thinking about how I should have acted differently in certain situations.*). Taken their item content into account, both of these items refer to rumination in recent or certain situations. Further edge related descriptive statistic and their related accuracy and stability estimates are in section 3.5 below.

Another aspect of inter-questionnaire activation are items activating neighbours from another subscales in the combined graph of rumination. Items such as *react* of the *Brooding* subscale (*Think 'Why do I always react this way?'*) are in-between of multiple other items belonging to different subscales. The item *react* can activate its direct neighbor *handle* (*Think 'Why can't I handle things better?'*) from the same subscale, but also, although in a less stable way, *analyze* of the *Reflection* subscale (*Analyze recent events to try to understand why you are depressed*) and *act around* of SCRS (*I criticize myself a lot for how I act around other people.*).

3.2 Internal Structure with an exponential random graph model

Our ERGM model was built with *Markov chain Monte Carlo* (MCMC) maximum likelihood estimation (1000 iterations) to obtain model parameters by generating samples from the space of possible networks allowed by the data.

Table 3: ERGM model fit

Term	Estimate	Std. Error	MCMC Error %	Statistic	p	p of EB
# of edges	-8.15	8.39	0	-0.97	0.33	0.00
GWESP [triangle]	3.53	5.00	0	0.71	0.48	0.97
Questionnaire (main effect)	-0.33	0.19	0	-1.74	0.08	0.42
Questionnaire (homophily effect)	-0.95	1.25	0	-0.76	0.45	0.28
Scale (homophily effect)	4.10	1.23	0	3.34	0.00	0.98

Note. MCMC=Monte Carlo Markov Chain; GWESP=Geometrically Weighted Edgewise Shared Partner Distribution; p of EB=probability of Edge Building; model built with Markov chain Monte Carlo Maximum Likelihood Estimation

An ANOVA analysis on the overall model indicates (with df=5, residual deviance=142.929, p < 0.001) that variables involved in the model explain the network connectivity in a highly non-trivial fashion. The detailed analysis reveals that *endogenous* effects such as the overall density of the network (number of *edges*) change the probability of the formation of an edge by 0), and at p<0.05 level the closed triangle formulation indicates a non-trivial impact on clustering (connectivity probability changed by 0.972). As expected, *exogenous* effects, such as two items belonging to the same sub-scale have a significant positive impact on connectivity (sub-scale changes edge probability by 0.984, keeping the rest of the network fixed). Questionnaires themselves have no impact on edge formation ('Questionnaire main effect'). In terms of goodness-of-fit, observed frequencies of degree levels differ in non-significant manner from *MCMC*-simulated expectations. In two cases of vertex degree (V_d=3 and V_d=6), the observed frequencies are above the simulated average (0 > the expected avg. of 1.28 for V_d=3; and 0 > the expected avg. of 0 for V_d=6), but differences are not significant.

Regarding the expected edgewise shared partners, the only deviation from expected levels is for 1 partner, where the observed frequency is 17 against 13.28. Additional MCMC diagnostics revealed good fit of the model, as the observed graph does not differ from the *MCMC* simulated graphs.

3.3 Association Network Inference

Results show low η_0 (0.051) indicating large mixing rate for the alternative hypothesis of edges having non-zero weights. The histogram of absolute correlation coefficients (see Figure 3. in Appendix) has a centered shape in contrast to the null component. The density-based *FDR* drops once empirical correlations reach at least ρ=0.1 (see the lowest section of Figure 3.).

3.4 Latent Network Modeling

Our next model was built with the *eigenmodel_mcmc* function of the R package *eigenmodel*. The function uses *Markov chain Monte Carlo* (MCMC) techniques to complete a model specification. By applying eigenvalue decomposition, we obtained the first two eigenvectors (see Figure 2) on the combined network (S=10000, number of iterations).

Our visual insight from Figure 2. is that the first eigenvector serves as a separator between *RRS* and *SCRS* having a posterior mean of $e\bar{v}_1$=1.344. The second eigenvector seems to separate between the upper half of the two-dimensional space of *SCRS* and the *brooding* scale, and the lower half being the *reflection* scale ($e\bar{v}_2$=2.332).

In order to test the goodness-of-fit of the eigenmodel, using *k*-fold cross-validation (k=5), we refitted the model to a subset of the original data, using the complement 1/5 fraction for predicting the elements of the adjacency matrix of our original network. We used three different models in this part as discussed above (Section **2.6**).

Summarizing the results from the *eigenmodels*, using the original combined network's adjacency matrix as a reference, we estimated the *Area Under the Curve* (AUC) statistics for each model. Compared to the latent model (AUC_{latent} =0.348), the inclusion of the questionnaires as a grouping factor resulted in a

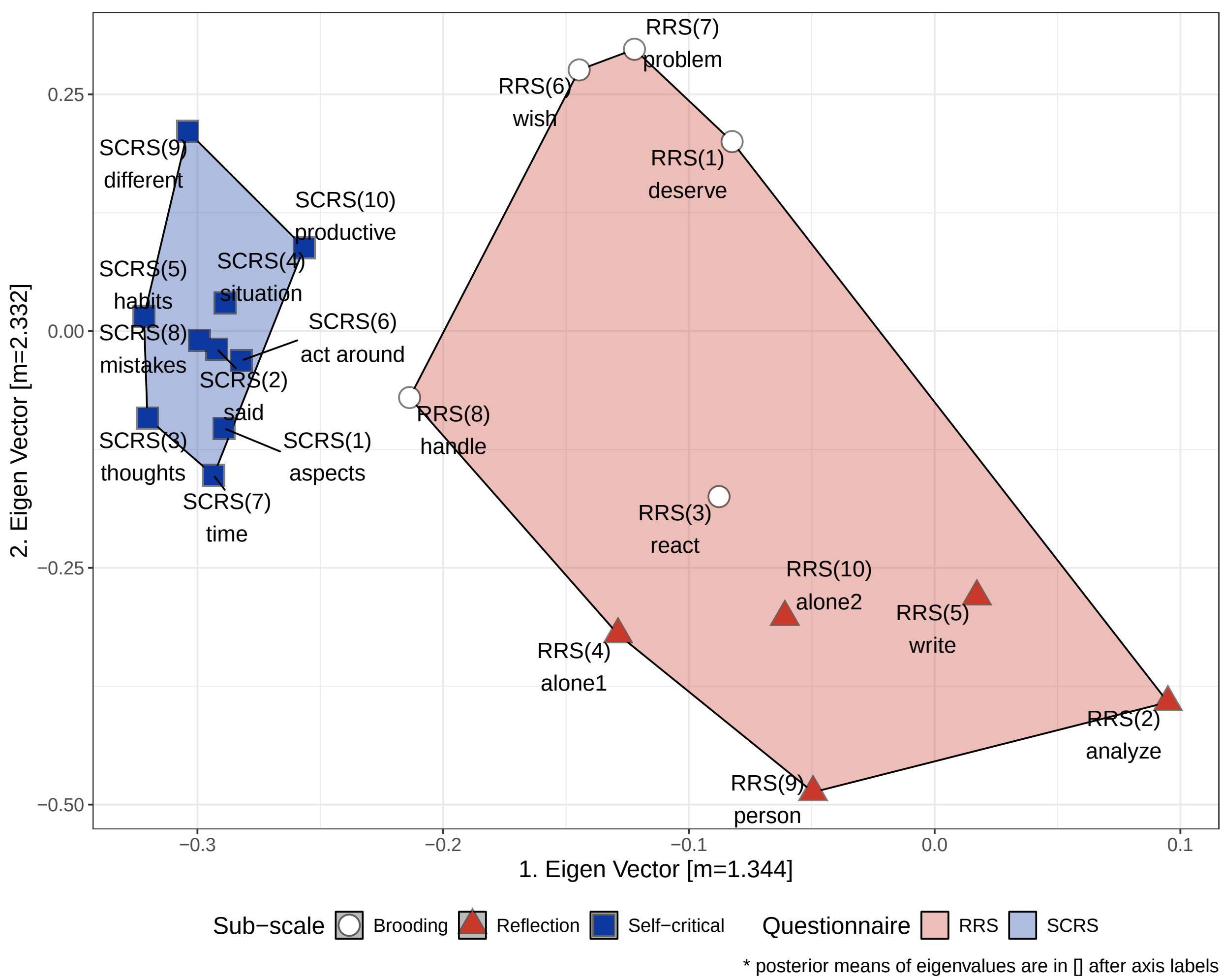

Figure 2: Vertices of the combined graph by the first two eigenvalues

better predictive model ($AUC_{latent+questionnaire}$ = 0.637). The final model involving the scale-level vertex characteristics has the best goodness-of-fit ($AUC_{latent+scale}$ =0.725). See Figure 4. in the Appendix for the corresponding ROC curves.

3.5 Power analysis, network accuracy and stability

We used the *bootnet* R package to have an estimate of edge-weight accuracy with *non-parametric bootstrap* (5000 times sampling). The resulting **accuracy** estimates have been visually incorporated in Figure 1 in the following manner. Edges are presented with dashed line for cases of the CI including 0.0 edge weights. For such cases, we cannot be sure if their weight persist given different samples and varying sample characteristics. Continuous edges represent stable edges according to the bootstrap estimate. Edges such as between the *SCRS(5)* and *SCRS(7)* items indicate moderate penalized partial correlation between items in the SCRS network (ρ=0.352), but the edge is presented with dashed line indicating lower stability. Examples for accurate and strong edges are *RRS(4)* (*alone1*; *"Go away by yourself and think about why you feel this way"*) and *RRS(10)* (*alone2*; *"Go someplace alone to think about your feelings"*). The edge weight for these items in their stand-alone *RRS* network is 0.572 (estimated edge-weight accuracy is with average 0.511 with lower and upper CI bounds of 0.454 - 0.567). These items exhibit the same pattern in the combined network with estimated edge-weight accuracy with average weight of 0.493; lower and upper CI bounds of 0.436 - 0.549.

In the SCRS network we can track a *path* (a graph movement [*walk*] without repeated edges or vertices) between *SCRS* 6, 5, 9 and 10 items (please refer to Table 5. for edge-level weight and accuracy estimates of *SCRS*).

Network **stability** estimates for our three networks were carried out by using the *bootnet* function (results in Table 3. below). We performed a *case dropping bootstrap* to simulate scenarios where fewer and fewer cases of the original sample would have been available and quantified the stability of the centrality metrics with the *Correlation Stability* (CS) coefficient. Bootstrap was performed with B=5000 samples.

We specifically examined how accuracy estimates of individual edges in their respective networks change compared to what we estimated in the combined graph of rumination. Although most of the stability and accuracy estimates decreased once entering into the combined graph, some increased, most notably the upper bounds of the 95% confidence intervals of accuracy for edges as *deserve - problem* and *analyze - alone1* of RRS and *situation - habits* and *different - time* of SCRS. Edge desriptive statistics as well as edge accuracy and stability statistics have been summarized for *RRS*, *SCRS* and their combined graphs in respective tables of Appendix (see Table 5., 6., and 7.).

As shown at (Epskamp, Borsboom, et al., 2017b), the *Correlation Stability* coefficent is not acceptable below 0.25, and preferably stays above 0.5.

Table 4: Correlation Stability coefficient estimates

Metric	RRS	RRS:Brooding	RRS:Reflection	SCRS	RRS & SCRS
Betweenness	0.01	0.26	0.10	0.07	0.04
Closeness	0.17	0.39	0.33	0.37	0.01
Edge Strength	0.28	0.56	0.75	0.28	0.52
Expected Influence	0.36	0.56	0.82	0.27	0.52
Edge Weight	0.70	0.58	0.85	0.72	0.66

We found satisfactory correlation stability of the Betweenness statistic of the Brooding (RRS) stand-alone network ($CS_{Brooding}$=0.26) as Brooding has retraceable communities, and as we already pointed out above, had the only Articulation Point of the RRS network. Brooding, Reflection and the SCRS networks have acceptable correlation statistics for the closeness statistic ($CS_{Brooding}$=0.39, $CS_{Reflection}$=0.33 and CS_{SCRS}=0.37). The Edge Strength (the sum of edge weight, in the present paper the partial correlation coefficients) and the Expected Influence statistics have good correlation stabilities on all networks; specifically, the Reflection stand-alone network exhibits high estimates for Edge Strength ($CS_{Reflection}$=0.75) and

Expected Influence ($CS_{Reflection}$=0.82) which might be driven by the high correlation between items 4. and 10. of RRS. The Edge Weight shows highly satisfactory patterns on all networks (*"Edge Weight"* row of Table 3.) which is the expected stability of our penalized partial correlation coefficients. For the combined network of RRS and SCRS, we see stable Edge Strength, Edge Weight and Expected Influence stability estimates, whereas Betweenness ($CS_{combined}$=0.04) and Closeness ($CS_{combined}$=0.01) are unsatisfactory. These findings suggest that the combined network tolerates very few sample loss and sample variance when it comes to the Betweenness centrality and the Closeness centrality measures, as all networks are densely connected and fewer items can stand out with higher local importance such as being between sub-networks.

4. Discussion

In the present paper, we examined the separation of two facets of rumination (a depressive dimension, lying along the *brooding-reflection* axis; and the *self-critical* aspects of rumination) using a network approach. Our findings suggest that the brooding and reflective aspects of depressive rumination and the self-critical rumination are three distinct, unique areas of the multifaceted construct of rumination. A network approach identified them as not merging with one-another, but establishing a stable and sensitive network of items. Thus, we provided further evidence that self-critical rumination is a separate construct from depressive rumination.

The **Self-Critical Rumination Scale** proved to have a one-factor structure. In its respective network, the **Ruminative Response Style Questionnaire** exhibits a two-factor structure. Our community detection estimates suggest though that the reflective aspects of rumination such as analyzing one's self and recent events for better emotional insights (items 2 and 9 of *RRS*: *"Analyze recent events to try to understand why you are depressed"* and *"Analyze your personality to try to understand why you are depressed"*) have more to common with brooding, or have a tendency to emerge as a third aspect. In analyzing the factor structure of *RRS* on samples from Argentina and the United States, the error variances of these two items were found to be correlated and thus ended up being excluded from the solution (Arana & Rice, 2017). The association of these items with depression, and therefore their elimination from any analysis related to Reflection have been suggested and practiced (Armey et al., 2009; Lee & Kim, 2014; Treynor, 2003). With concurring results on these items, others suggest that due the confounding effect of these items with depression or other sampling characteristics, the boundary of *reflection* and *brooding* can become less strict (Whitmer & Gotlib, 2011). On a sample of Dutch students, researchers excluded these two items for reasons of their "*reference to negative feelings*", and the resulting three-item Reflection scale was names as "*reflective self-regulation*" (Brose et al., 2020). Although, in a recent study on samples of Peruvian students, the separation of the "analyze" items was not present (Valencia & Paredes-Angeles, 2022). A clear two-factor structure of RRS was revealed on Columbian samples (Ruiz, 2017) or on Japanese sample (Hasegawa, 2013). These findings on two- or three-factor structures of RRS support the notion of our findings that the *"analyze"* items are sensitive to sampling characteristics, demographics, and as we have seen in our combined network, to covariates or additional activation paths taken into account. Items #2 and #9 of *RRS* on their own graph emerged as a third component, presented a low *true positive rate* of 0.22 each, but this pattern dissolved in a combined graph with the self-critical aspects of rumination accounted for.

Separation between scales in the **combined graph** is underlined by our *eigenmodel*, as in the two-dimensional space of the first two eigenvectors, expected grouping of *RRS* and *SCRS* items can be achieved. Items #2 and #9 of RRS are well separated from the bulk of the rest of the RRS items, supporting our above conclusion of them being a part of reflection or found a new component. Brooding has more associations with the self-critical aspects of rumination than reflection, which is highlighted by our eigenvector analysis as well as the sparse, unstable and weak edges with which reflection connects to brooding and self-critical rumination in their combined network. Our graphs suggests and support the notion of not referring to a psychological attribute as a sum score, but rather take into account the underlying activation pattern (Bernstein et al., 2019; Bork et al., 2019). Items modelled as nodes of a network present activation patterns inter-subscales, an example of which in our research is the interplay between items *react* and *handle* (both from Brooding), *analyze* (Reflection) and *act around* (SCRS). This example shows how the generic rumination on acting and the reasons of reactions activate other ruminative aspects, such as analyzing and understanding one's actions, and increased self-criticism in a social context. In a network analysis, Williams et al. (2021) combined items of *"analyze"* into one node.

We have gathered evidence of scale playing an important role in the edge-building process. According to our exponential random graph model, the internal scales of the combined network indeed have a statistically significant impact on associations between items, whereas the effect of questionnaire dissolved, no longer having an impact on the probability of edge formation. Edge descriptive statistics as well as edge accuracy and stability statistics have been summarized for *RRS*, *SCRS* and their combined graphs in respective tables of Appendix (see Table 5., 6., and 7.). This suggest that all three scales are effective in assessing a separate facet of rumination when used together. As shown, associations within the combined network only have low chances of false discovery.

Results indicate that network associations are accurately aligned to the internal factor structure, while some associations are more sensitive to sampling characteristics and sampling variation. Edge strengths and edge weights have also high stability and relatively high resistance to sample loss, whereas betweenness is remarkably sensitive to case-dropping. This suggest that on different samples with lesser count of observations and varying sample characteristics, the centrality of a node conceptualized as being a connecting point between subnetworks (i.e. betweenness) can be less stable.

5. Limitations

Our results have provided a view on rumination as an interplay between depressive rumination and self-criticism, but no other construct was involved in the present study. By including depression, anxiety or social aspects of the individual the combined view on rumination could be further expanded. Regarding our sample characteristics, the present results are based on an adolescent, non-clinical sample. Future studies with the same design, but on clinical samples such as subjects with high levels of depression, anxiety, self-harm, assumed to also have high self-criticism, would allow a broader and more contrasted view on the matter. The inclusion of different age groups will also broaden our view on the field.

6. Conclusions

Network analysis is an evolving and in research psychology, an emerging approach to draw *psychological networks*, and eventually confirm or question existing findings on confirmatory analysis. In the present paper, scrutinizing rumination as patterns of activation among its measuring items, we provided a view on depressive and self-critical rumination as a common network. Being in their respective, isolated network, depressive rumination and self-critical rumination confirmed their structures (both solid and blurred, like the "analyze" items) known to literature. The combined network of rumination presented less-to-none blurred items, and has drawn a clear picture of a large network where activation on nodes can present rumination as a pattern rather then a summed score.

Future steps include analyzing such data from a person-level graph perspective, where vertices are individuals, and edges can be some distance measure between vertices based upon demographics and survey data. Also, if keeping our analysis on the item-level, as in the present paper, we would extend our methodology to directed acyclic graphs (DAGs) which would allow causal assumptions to support our understanding various facets of rumination.

7. Declarations

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Acknowledgements

We thank for all the BA students who helped us in data collection. Conflicts of interest/Competing interests. The authors declare that they have no competing interests.

Ethics approval

The protocol was designed in accordance with the Declaration of Helsinki and the study was ethically approved by the Institutional Review Board of Eötvös Loránd University, Faculty of Education and Psychology.

Consent to participate

Participation in the study was voluntary and anonymous for the participants (high school students). Informed consent was obtained from the participants and their parents. School principals provided institutional consent.

Consent for publication

Not applicable.

Availability of data and materials (data transparency)

https://github.com/TamasSmahajcsikszabo/rumination_lasso/data/

Code availability (software application or custom code)

A companion R package 'NetworkAnalysisTools': https://github.com/TamasSmahajcsikszabo/networkAnalysisTools
Codes for the article: TBD

Authors' contributions

GK and ZR conceptualized the study design and supervised the data collection. TSSZ performed data analysis and GK, NK and RU contributed to the interpretation of the results. TSSZ and NK drafted the manuscript. GK, ZR, NK, RU reviewed the first draft and provided critical feedback. All authors read and approved the final manuscript.

Appendix

Table 5: Network descriptive statistics for RRS, SCRS and their combined networks

	1.	2.	3.	4.
Correlations:				
1. RRS				
2. RRS - brooding	0.82			
3. RRS - reflection	0.85	0.40		
4. SCRS	0.64	0.63	0.44	
Female Mean (SD)	21.59 (4.85)	10.88 (2.77)	10.71 (3.08)	25.19 (7.15)
Male Mean (SD)	19.07 (4.59)	9.65 (2.72)	9.42 (2.80)	21.23 (6.71)
Cronbach's α	0.75	0.65	0.70	0.89
Cohen's d (20% trimmed)	0.61	0.54	0.47	0.53
Cliff's delta	P(f>m)=0.64	P(f>m)=0.59	P(f>m)=0.58	P(f>m)=0.62

Deg.=Degree; Str.=Strength; Bet.=Betweenness; Clo.=Closeness; EI1= One-step Expected Influence; EI2= Two-step Expected Influence; Br.Str.=Bridge Strength; Br.Bet.=Bridge Betweenness; Br.Cl.=Bridge Closeness; Br.EI1=One-step Bridge Expected Influence; Br.EI2=Two-step Bridge Expected Influence; TPR=average True Positive Rate

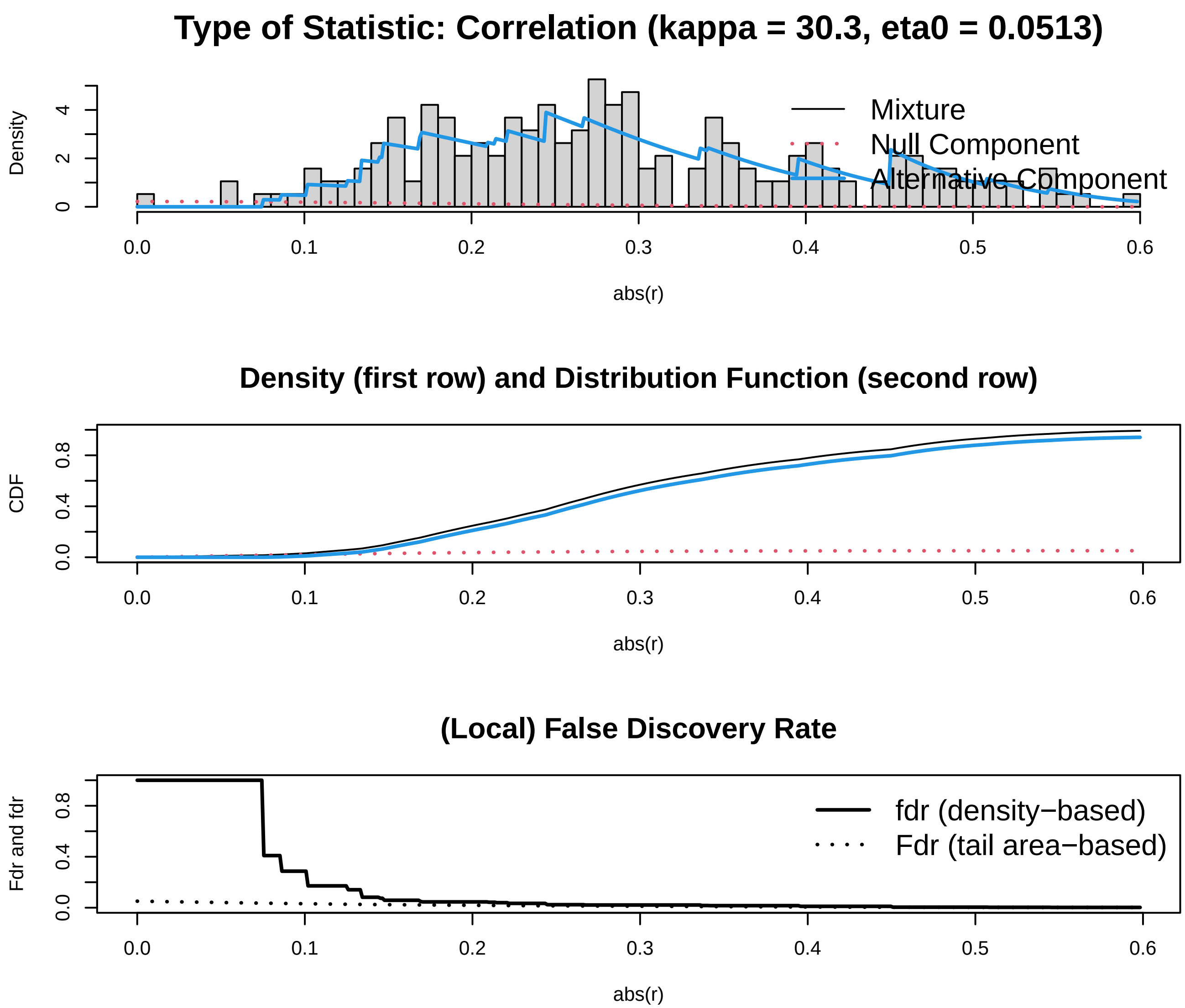

Figure 3: False Discovery Rate (FDR) for the combined network

Table 6: Edge Accuracy and Stability Statistics for the RRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.%:0	Miss.%:0.2	Miss.%:0.4	Miss.%:0.6	Miss.%:0.7
RRS (1)	RRS (7)	0.20	0.13	0.27	0.20	0.20	0.19	0.18	0.12
RRS (2)	RRS (4)	0.15	0.08	0.22	0.16	0.15	0.14	0.12	0.08
RRS (2)	RRS (8)	0.00	-0.01	0.01	0.00	0.00	0.00	0.00	0.00
RRS (2)	RRS (9)	0.25	0.19	0.32	0.25	0.25	0.26	0.24	0.22
RRS (3)	RRS (7)	0.06	-0.07	0.18	0.01	0.01	0.02	0.02	0.02
RRS (3)	RRS (8)	0.23	0.16	0.30	0.23	0.23	0.23	0.22	0.17
RRS (3)	RRS (9)	0.09	-0.04	0.22	0.10	0.07	0.05	0.04	0.02
RRS (4)	RRS (10)	0.51	0.45	0.57	0.51	0.51	0.51	0.51	0.50
RRS (4)	RRS (8)	0.06	-0.05	0.17	0.06	0.04	0.02	0.02	0.02
RRS (4)	RRS (9)	0.09	-0.03	0.20	0.10	0.08	0.05	0.05	0.03
RRS (5)	RRS (10)	0.11	0.00	0.22	0.12	0.11	0.07	0.06	0.03
RRS (5)	RRS (8)	0.00	-0.04	0.04	0.00	0.00	0.00	0.00	0.00
RRS (6)	RRS (7)	0.20	0.13	0.27	0.20	0.20	0.19	0.17	0.10
RRS (6)	RRS (8)	0.09	-0.04	0.22	0.11	0.07	0.05	0.03	0.02
RRS (7)	RRS (9)	0.00	-0.01	0.01	0.00	0.00	0.00	0.00	0.00
RRS (8)	RRS (10)	0.00	-0.03	0.04	0.00	0.00	0.00	0.00	0.00
RRS (8)	RRS (9)	0.09	-0.03	0.22	0.11	0.09	0.05	0.04	0.03

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to;Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Miss.%=Estimated average stability at the given missing rate of sample size

Table 7: Edge Accuracy and Stability Statistics for the SCRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.%:0	Miss.%:0.2	Miss.%:0.4	Miss.%:0.6	Miss.%:0.7
SCRS (1)	SCRS (5)	0.16	0.09	0.23	0.16	0.16	0.16	0.15	0.13
SCRS (1)	SCRS (8)	0.04	-0.06	0.13	0.00	0.02	0.02	0.01	0.02
SCRS (2)	SCRS (3)	0.03	-0.06	0.11	0.00	0.01	0.02	0.01	0.02
SCRS (2)	SCRS (5)	0.00	-0.03	0.03	0.00	0.00	0.00	0.00	0.00
SCRS (2)	SCRS (8)	0.18	0.11	0.25	0.18	0.18	0.18	0.17	0.15
SCRS (3)	SCRS (10)	0.18	0.11	0.24	0.18	0.18	0.18	0.17	0.14
SCRS (3)	SCRS (5)	0.11	0.03	0.19	0.11	0.11	0.10	0.09	0.07
SCRS (3)	SCRS (7)	0.25	0.18	0.32	0.25	0.25	0.25	0.25	0.25
SCRS (4)	SCRS (5)	0.11	0.02	0.20	0.12	0.11	0.10	0.09	0.07
SCRS (4)	SCRS (8)	0.23	0.16	0.29	0.23	0.23	0.23	0.23	0.22
SCRS (5)	SCRS (6)	0.29	0.23	0.36	0.30	0.30	0.29	0.30	0.29
SCRS (5)	SCRS (7)	0.06	-0.05	0.16	0.04	0.04	0.04	0.04	0.04
SCRS (5)	SCRS (9)	0.20	0.13	0.27	0.20	0.20	0.20	0.19	0.19
SCRS (6)	SCRS (8)	0.09	-0.02	0.19	0.10	0.08	0.06	0.05	0.05
SCRS (8)	SCRS (9)	0.12	0.03	0.20	0.12	0.12	0.11	0.08	0.05
SCRS (9)	SCRS (10)	0.30	0.23	0.36	0.30	0.30	0.30	0.29	0.29

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to;Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Miss.%=Estimated average stability at the given missing rate of sample size

Table 8: Edge Accuracy and Stability Statistics for the RRS and SCRS network

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.%:0	Miss.%:0.2	Miss.%:0.3	Miss.%:0.6	Miss.%:0.7
RRS (1)	RRS (6)	0.06	-0.09	0.22	0.00	0.01	0.01	0.00	0.01
RRS (1)	RRS (7)	0.13	-0.03	0.29	0.16	0.11	0.07	0.03	0.01
RRS (2)	RRS (3)	0.06	-0.09	0.21	0.00	0.01	0.01	0.01	0.01
RRS (2)	RRS (4)	0.13	0.01	0.26	0.15	0.13	0.10	0.04	0.02
RRS (2)	RRS (9)	0.24	0.17	0.30	0.24	0.24	0.24	0.19	0.11
RRS (3)	RRS (8)	0.16	0.04	0.28	0.17	0.16	0.13	0.06	0.02
RRS (3)	SCRS (6)	0.03	-0.08	0.14	0.00	0.00	0.00	0.00	0.00
RRS (4)	RRS (10)	0.49	0.44	0.55	0.49	0.49	0.49	0.49	0.46
RRS (4)	RRS (9)	0.06	-0.08	0.19	0.01	0.02	0.01	0.02	0.01
RRS (4)	SCRS (9)	0.02	-0.07	0.10	0.00	0.00	0.00	0.00	0.00
RRS (5)	RRS (10)	0.06	-0.08	0.21	0.01	0.01	0.01	0.00	0.00
RRS (5)	RRS (9)	0.19	0.09	0.29	0.20	0.19	0.17	0.08	0.03
RRS (6)	RRS (7)	0.14	0.00	0.28	0.16	0.13	0.09	0.04	0.02
RRS (6)	SCRS (4)	0.17	0.08	0.26	0.17	0.17	0.15	0.09	0.05
RRS (7)	RRS (8)	0.06	-0.08	0.21	0.00	0.01	0.01	0.01	0.00
RRS (7)	SCRS (5)	0.07	-0.06	0.21	0.03	0.03	0.03	0.02	0.01
RRS (7)	SCRS (9)	0.12	-0.01	0.25	0.14	0.11	0.07	0.05	0.04
RRS (8)	SCRS (10)	0.12	-0.01	0.25	0.15	0.12	0.09	0.04	0.03
RRS (9)	SCRS (6)	0.02	-0.07	0.12	0.00	0.00	0.00	0.00	0.00
SCRS (1)	SCRS (2)	0.24	0.17	0.30	0.24	0.24	0.24	0.23	0.18
SCRS (1)	SCRS (3)	0.14	0.05	0.23	0.15	0.15	0.13	0.09	0.05
SCRS (1)	SCRS (5)	0.15	0.06	0.24	0.16	0.15	0.14	0.11	0.06
SCRS (1)	SCRS (7)	0.05	-0.07	0.17	0.01	0.01	0.02	0.01	0.01
SCRS (2)	SCRS (4)	0.24	0.18	0.31	0.24	0.24	0.24	0.24	0.19
SCRS (2)	SCRS (6)	0.07	-0.05	0.19	0.07	0.04	0.03	0.01	0.02
SCRS (2)	SCRS (8)	0.16	0.08	0.24	0.16	0.16	0.15	0.11	0.07
SCRS (3)	SCRS (10)	0.15	0.05	0.24	0.16	0.16	0.14	0.10	0.06
SCRS (3)	SCRS (4)	0.17	0.10	0.25	0.17	0.17	0.17	0.14	0.09
SCRS (3)	SCRS (5)	0.08	-0.04	0.20	0.09	0.06	0.05	0.03	0.02
SCRS (3)	SCRS (7)	0.23	0.17	0.30	0.24	0.24	0.23	0.23	0.16
SCRS (4)	SCRS (5)	0.11	0.00	0.22	0.12	0.11	0.09	0.06	0.04
SCRS (4)	SCRS (6)	0.03	-0.07	0.14	0.00	0.01	0.01	0.01	0.01
SCRS (4)	SCRS (8)	0.20	0.13	0.27	0.20	0.20	0.20	0.18	0.13
SCRS (5)	SCRS (6)	0.29	0.22	0.36	0.29	0.29	0.29	0.28	0.25
SCRS (5)	SCRS (9)	0.17	0.10	0.25	0.17	0.17	0.17	0.15	0.10

Vx.1	Vx.2	Avg. Acc.	L.B.Acc.	U.B.Acc.	Miss.%:0	Miss.%:0.2	Miss.%:0.3	Miss.%:0.6	Miss.%:0.7
SCRS (6)	SCRS (7)	0.17	0.09	0.25	0.17	0.17	0.17	0.12	0.08
SCRS (7)	SCRS (9)	0.14	0.03	0.24	0.14	0.14	0.12	0.09	0.06
SCRS (8)	SCRS (9)	0.09	-0.03	0.21	0.12	0.09	0.06	0.03	0.02
SCRS (9)	SCRS (10)	0.26	0.20	0.33	0.27	0.27	0.27	0.27	0.21

Vx.1=1st Vertex the edge attaches to; Vx.2=2nd Vertex the edge attaches to;Avg.Acc.=Average Edge Weight by Bootstrap; L.B.Acc.=Lower Bound of 95% CI for bootstrapped network accuracy; U.P.Acc.=Upper Bound of 95% CI for bootstrapped network accuracy; Statistic=type of network characteristic; Miss.%=Estimated average stability at the given missing rate of sample size

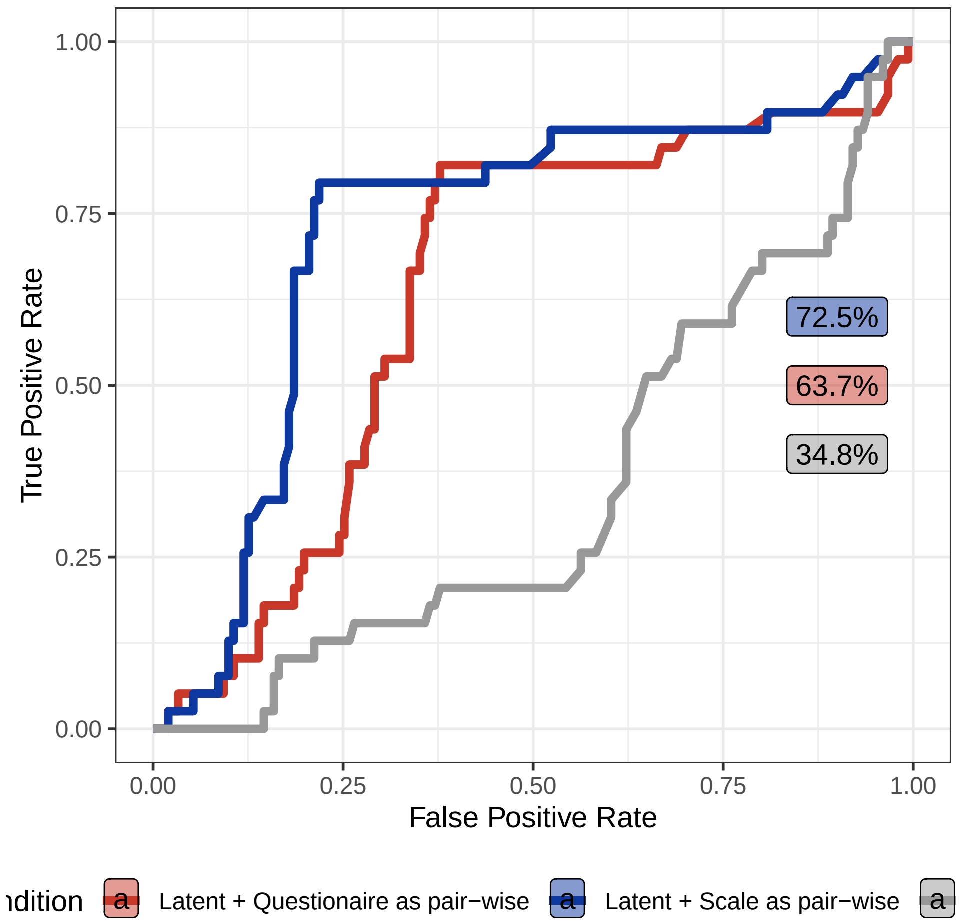

Figure 4: ROC curves of the combined RRS-SCRS network

Table 9: Glossary of network analysis relevant terms mentioned in the article

Term	Definition
Articulation point	A measure of vertex influence: V(i) vertex of G graph is an Articulation Point, if removed from the graph, would segment, cut or disintegrate the graph; therefore is considered locally important.
Association network	A network where neighbouring vertices when connected are assumed to being associated; in other words, edges represent some measure of correlation or other kind of association; this makes them different from energy or information flow networks, for instance.
Community Detection	A group of algorithmic methods to find clusters of nodes whose community can be described by systematic variance encoded in data represented in edges and node characteristics, and therefore are non-trivial. In the current paper, we used four selected algorithms to estimate how well they identify the original subscale-structure of the RRS and SCRS questionnaires.
Cut vertex	see ‘Articulation point’
Edge	A link (E, arc) between nodes of a network. Undirected if direction of the connection (e.g. flow of energy or information) is not essential to the network; directed (uni- or bidirectional) otherwise. Can be weighted by some measure of association, frequency or other attribute.
Exponential Random Graph Models	ERGM; a family of models to the analogy of generalized linear models adopted to network modeling. They allow the construction, fitting and comparison of network models. Their nature is structural, incorporating different configurations of a graph (sets of possible edges). ERGMs have numerous extensions and variations.
Graph	see ‘Network’

Term	Definition
Markov Chain Monte Carlo	Monte Carlo methods are a collection of techniques which approximate a quantity (mean, variance, some other measure of location, for instance) by generating random values of a random variable. In the present paper, MCMC was used to get the ERGM model parameters; and also in the fitting of the eigenmodel for the latent network modeling.
Network	Graph G = (V,E), a construct of V vertices and E edges.
Network, Adjancency matrix	Given v number of vertices, the adjacency matrix is a v X v matrix, where any [i,j] cell's value reflects with 0 or 1 whether a selected pair of vertices are connected or not
Network, Clustering coefficient	Transitivity, a descriptive measure to reflect the degree of connectedness [or process of clustering] in a graph by giving the ratio of connected triangles (triads of vertices) versus the total number of triangles in the graph.
Network, Diameter	The value of the longest distance (traceable route along edges and incident nodes) in a graph.
Network, Order	Descriptive; order of a network is the number of vertices.
Network, Size	Descriptive; size of a network is the total number of edges.
Network, Vertex, Betweenness	Descriptive of vertex centrality; quantifies how much a given vertex is between other pairs of vertices, thus has a connecting power.
Network, Vertex, Closeness	Descriptive of vertex centrality; assumes that important vertices are closer to others; its measure is inverse of total distances to other vertices.
Network, Vertex, Degree	Descriptive; the degree of a vertex is the number of edges incident on the given vertex. Vertex degrees are summarized as Degree Distribution of a graph.
Network, Vertex, Strength	Descriptive of vertex centrality; equals the sum of edge weights incident upon the given vertex, reflecting vertex centrality (importance). Vertex Strength Distribution is a histogram of vertex strength estimates of an entire graph.

Term	Definition
Node	Nodes (V,vertices) are representation of constructs, latent variables, subjects or any level of attributes in a network. Like edges, they can inherently bear any attributes.
Penalization	In regression, penalization introduces additional penalty to the sum of squared errors the model is trying to minimize. This way, by allowing a small increase in bias, the model is expected to exhibit a lower level of variance. The graphical LASSO method, used in the current paper, uses an absolute fraction of coefficietnts as penalties. With the applied penalty, insignificant or weak associations are expected to shrink towards 0.

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